

AZURE ARCHITECTURE

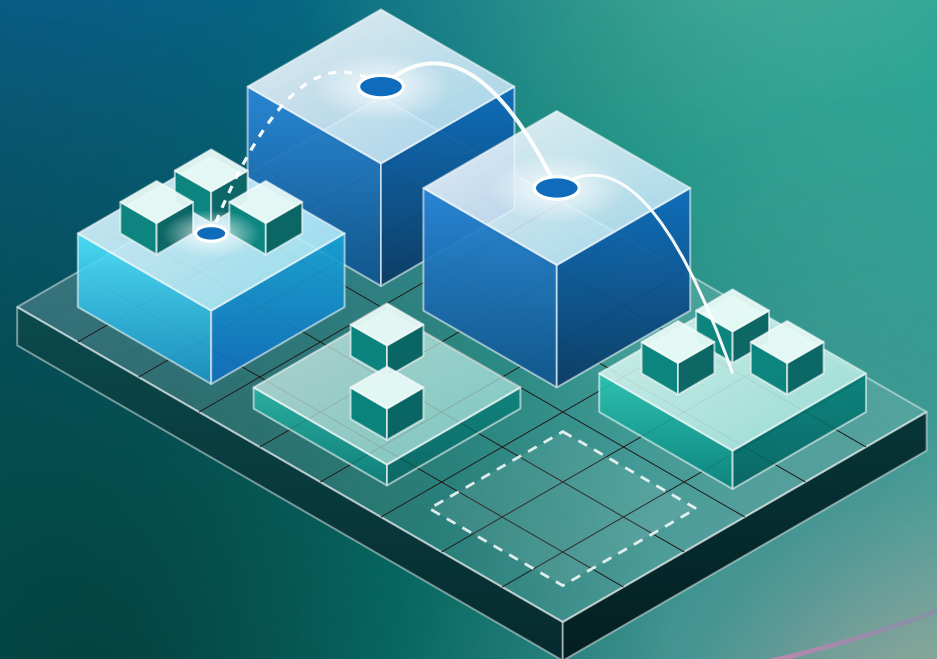
EXECUTIVE BRIEF

The adaptable **multi-region** Azure platform

A leadership summary: how to make additional Azure regions governed options — adopted when business and workload requirements justify them, without duplicating the platform in every region.

THREE PLANNING CONSTRUCTS

- Regional Platform Connectivity Profiles
- Workload Landing Zone Archetypes
- Regional Qualification



EXECUTIVE SUMMARY

Evolve landing zones from fixed regional footprints into a governed operating model

Adding an Azure region shouldn't mean starting another landing-zone project. In a governed geographic operating model — for example, across Europe — **regional placement becomes an adaptable, application-level decision** instead of a permanent architectural dependency.

FIVE MOVES

01

**Standardize governance**

Policies, security controls, operational rigor, and compliance are the constants across the geography.

02

**Decouple connectivity**

Plan resilient hybrid connectivity to Azure geographically, not again for every region.

03

**Tier the regional footprint**

A small number of strategic hubs carry the full stack; other regions use more efficient patterns.

04

**Classify workloads by archetype**

Four Workload Landing Zone archetypes describe what each workload needs from the platform.

05

**Preserve regional optionality**

Qualified regions stay available for the next workload when capacity, services, or regulation change.

Adaptability becomes a **deliberate architectural property** rather than a reaction to the next country-level power, regulatory, or supply-chain constraint.

Standardize the operating model, not necessarily the regional footprint.

THE GOAL

The goal is not to select the perfect two regions. **It is to make additional regions significantly easier to adopt when business and workload requirements justify them.**

REFRAME THE OBJECTIVE

01 An operating model that keeps working as the environment changes

The architecture should not depend on the regions we pick today staying the right answer.

Azure landing zones have usually been established in a particular region — often a primary and a disaster-recovery region — with a fairly complete set of networking, security, management, and shared services around them. That works well while the regions an organization wants to use stay relatively static.

The challenge now is different. Regulatory requirements and power availability can change across European countries, even as Microsoft continues to invest in and grow capacity across Europe. For regulated organizations, the question becomes how to **de-risk continued growth** in an industry-wide dynamic that keeps changing.

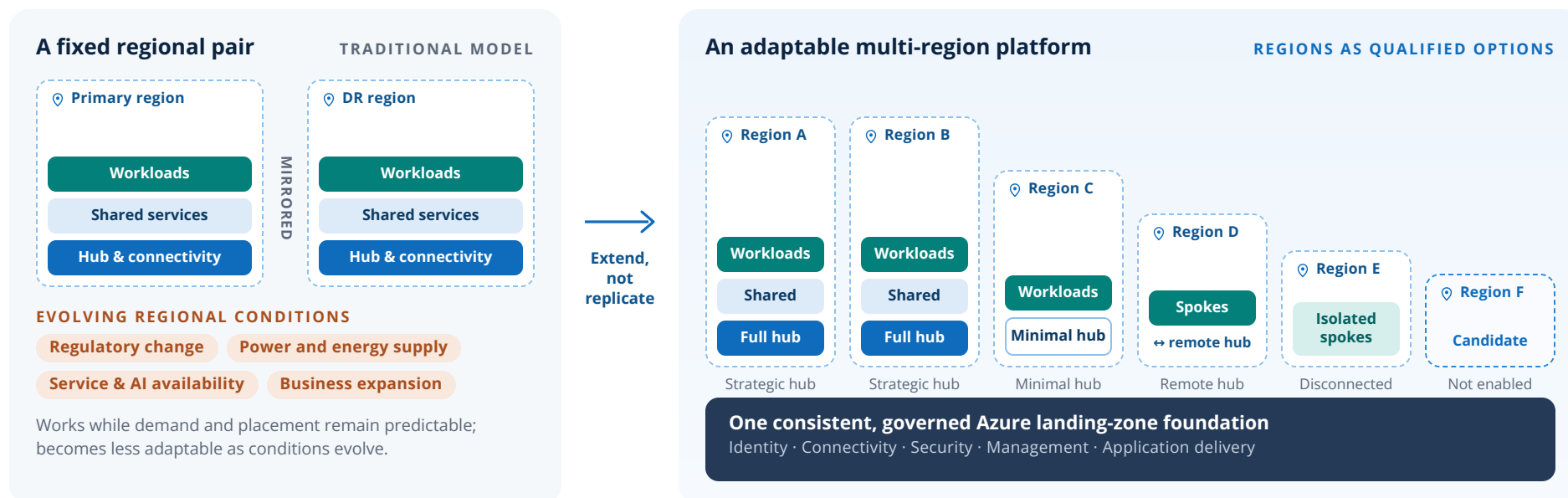


FIGURE 1 From a fixed regional pair to governed regional options. Instead of mirroring a complete footprint in each new region, the platform extends one governed foundation and gives each region only the capability its workloads require. Regional capability is a spectrum, not a copy.

FROM A REGION MINDSET TO A GEOGRAPHY MINDSET

02 Define the landing zone by geography, not by region

Broaden the architectural boundary. The individual regions underneath become adaptable deployment locations.

Rather than defining the cloud architecture around a small number of permanently designated Azure regions, define the overarching landing zone in geographic terms — for a European estate, a boundary such as the EU Data Boundary, governed by common security, operational, and compliance policies across multiple regions.

At that level, policies, security controls, operational rigor, and compliance requirements are the constants. The Azure regions underneath become qualified deployment options, and regional implementation adapts only where business, workload, or regulatory requirements call for it.

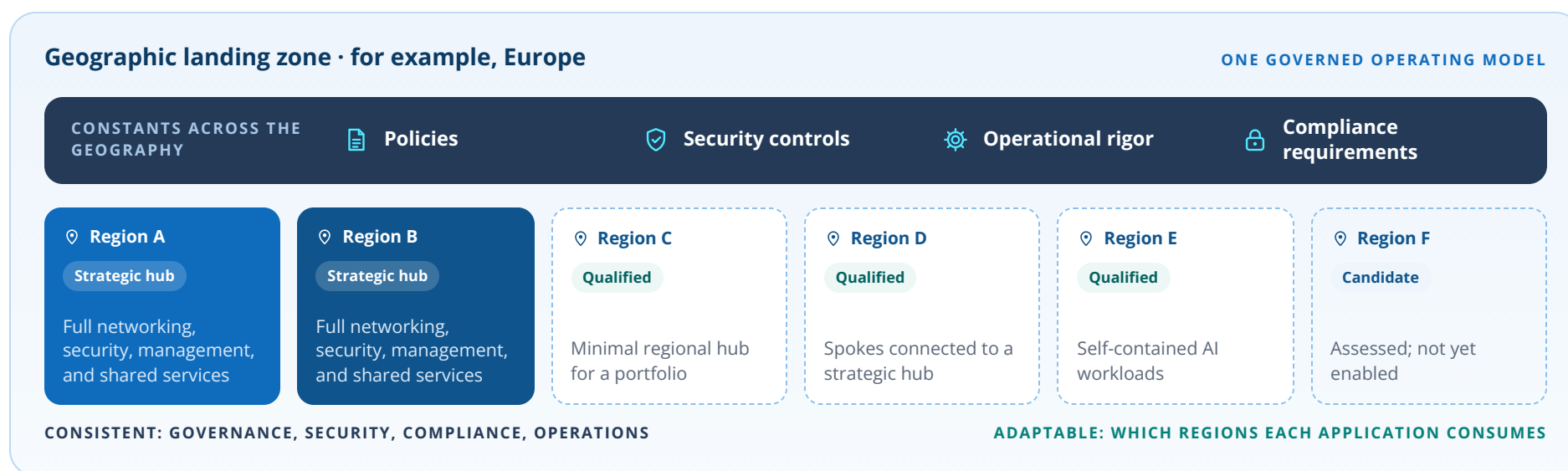


FIGURE 2 A geographic landing zone. The operating model is standardized across the geography; regions underneath are adaptable deployment locations with only the capabilities their workloads need. Region names and roles are illustrative.



Standardize the operating model, not necessarily the regional footprint.

DECOUPLE CONNECTIVITY FROM INDIVIDUAL REGIONS

03 Plan hybrid connectivity once, for the geography

Instead of rebuilding the full networking stack every time another Azure region is adopted, treat connectivity as a geographic design.

Establish redundant connectivity into the Azure backbone through strategically located ExpressRoute or ExpressRoute Direct peering locations. That provides route diversity and lets connectivity be planned for the geography rather than independently for every region. **Enabling an additional Azure region does not by itself require another ExpressRoute circuit;** add circuits, peering locations, gateways, or paths only where reachability, diversity, failure-domain isolation, latency, bandwidth, routing, or resiliency requirements justify them.

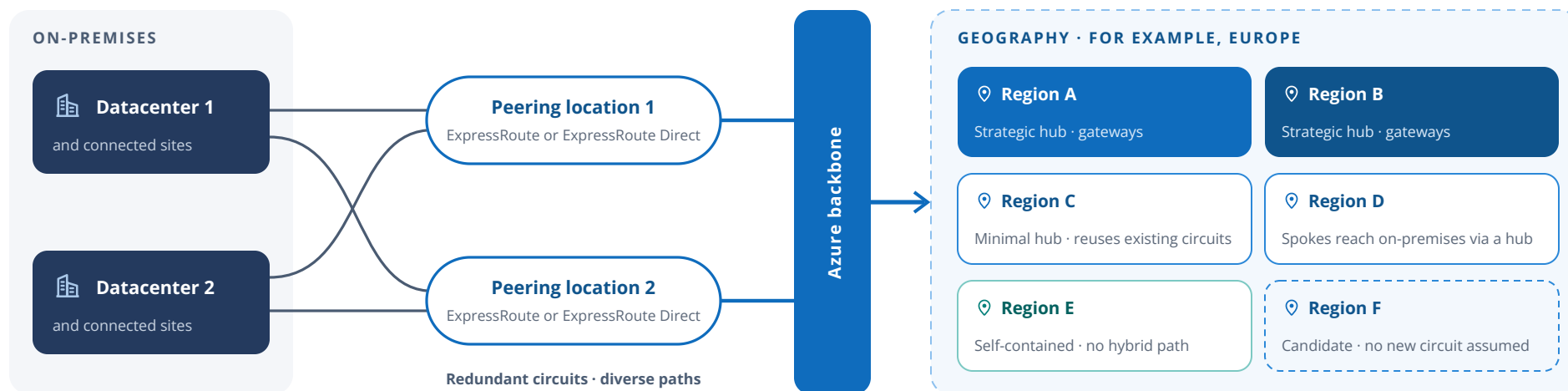


FIGURE 3 Geographic hybrid connectivity. Diverse paths through strategically located peering locations reach the Azure backbone once; regions consume that foundation through their hubs or a strategic hub. Reachability depends on circuit type — for example, Standard circuits reach regions in the same geopolitical region, Premium extends globally — so validate it per design. Locations and roles are illustrative.



Route diversity

Redundant circuits across separate peering locations remove single points of failure for the whole geography.



Decoupled from each region

Hybrid connectivity, hub placement, and application connectivity are separate decisions with separate tradeoffs.



Sustainable economics

A complete footprint in every additional region is expensive and hard to sustain; reuse what already meets the requirements.

04

TIER THE REGIONAL FOOTPRINT

Not every region needs a full landing zone

A few strategic hubs carry the full stack; other regions use the most efficient pattern that meets their requirements.

With a geographic foundation in place, the full landing zone does not have to be copied into every Azure region. Maintain a **small number of strategic geographic hubs** — often two or three regions — with the full networking, security, management, and shared-service footprint.

Other regions are consumed through more efficient patterns appropriate to each workload, evaluated application by application. The four connectivity profiles are **not maturity stages**: adoption growth alone does not justify moving toward a Full Regional Hub.

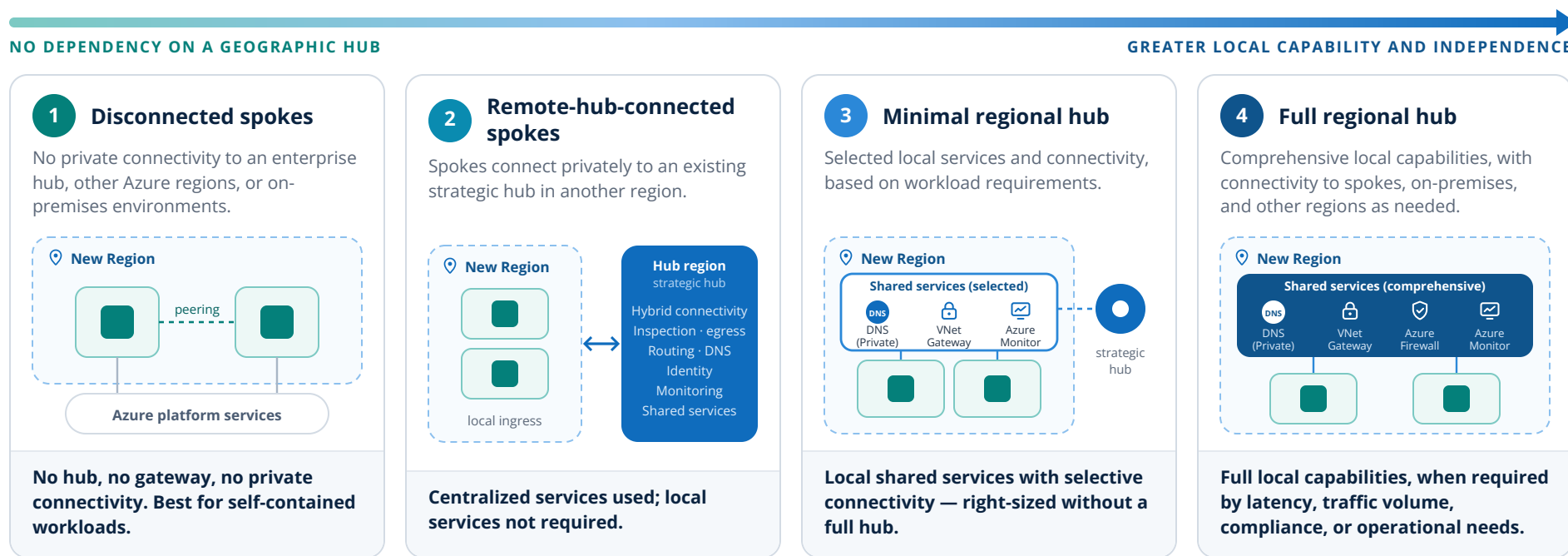


FIGURE 4 The spectrum of regional platform connectivity profiles. Moving right adds local capability and independence — and cost and operational scope. Services shown are examples; each hub includes only what its workloads require. Any profile can be the right long-term target.

04 · TIER THE REGIONAL FOOTPRINT

Pay for local capability only where it is justified

How the profiles trade local capability for platform scope

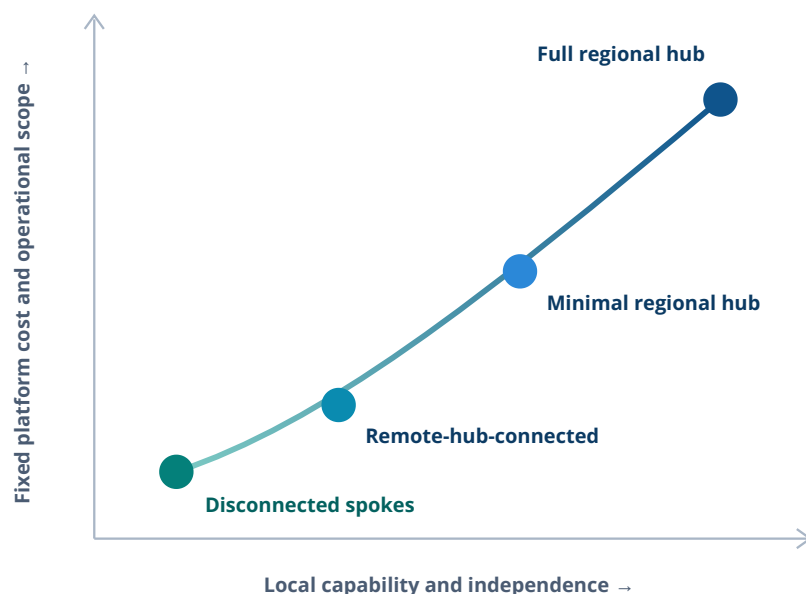


FIGURE 5 A qualitative trade-off, not a price list. More local capability usually raises fixed cost and operational scope, but can reduce cross-region dependency, latency exposure, and availability coupling.

There is no required progression between profiles. Select the **most efficient profile** for current requirements; different profiles coexist across the estate under the same landing-zone guardrails.

Profiles coexisting across one estate

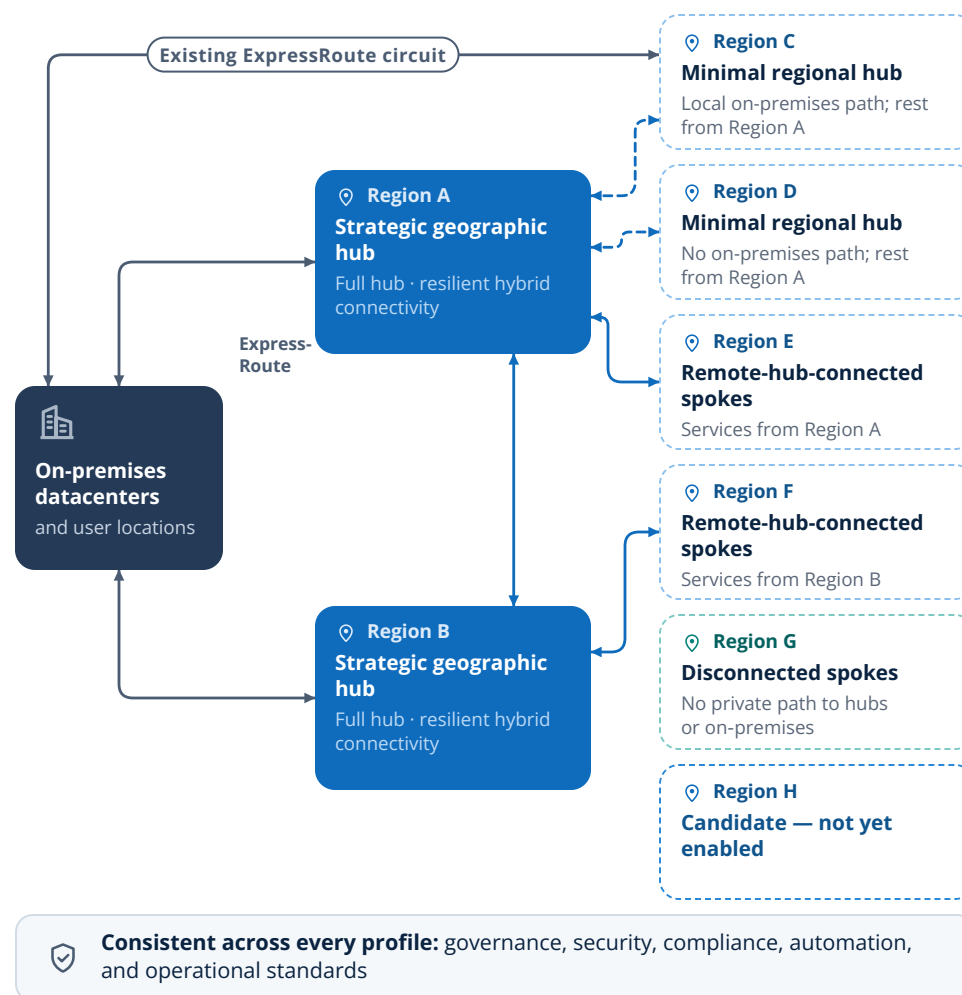
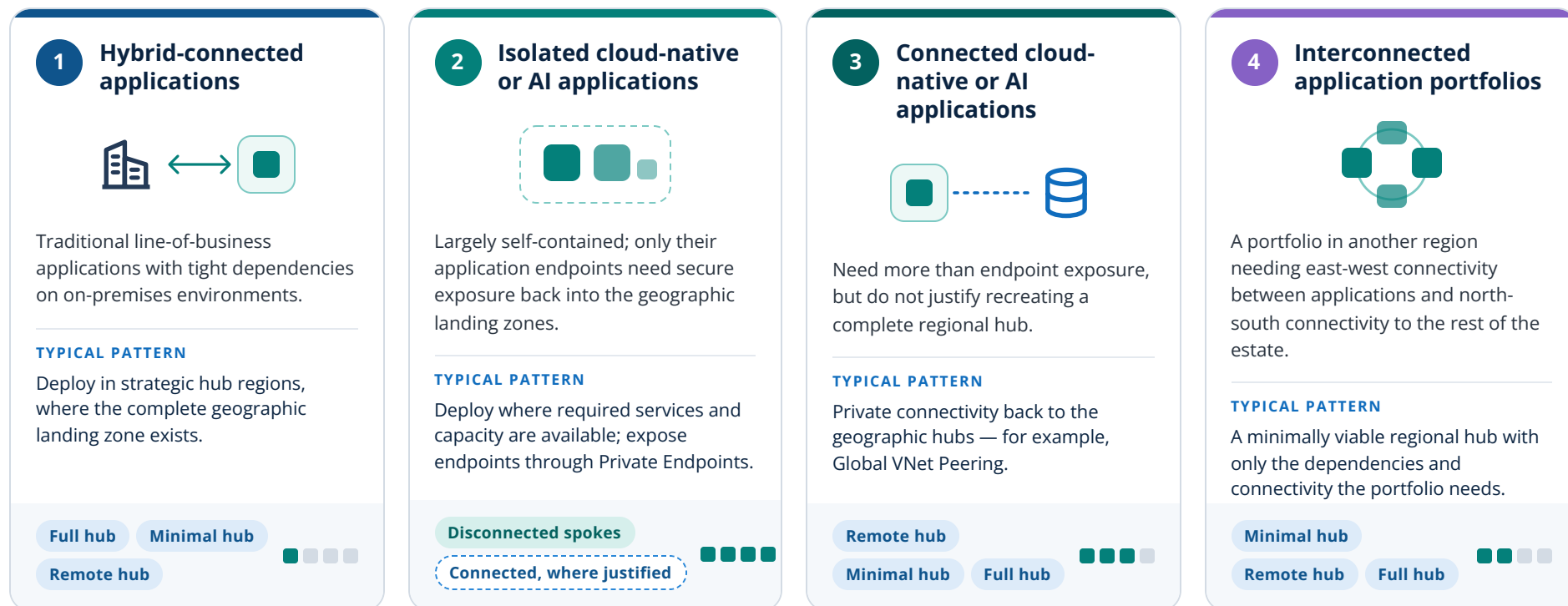


FIGURE 6 An illustrative estate: two strategic hubs provide resilient hybrid connectivity; other regions consume what they need. Minimal hubs take selected services from Region A (dashed) — Region C also reaches on-premises by reusing an existing ExpressRoute circuit, Region D does not. Both are valid. Names and placements are examples only.

CLASSIFY WORKLOADS BY ARCHETYPE

05 Four Workload Landing Zone archetypes

Archetypes describe **what applications require**; connectivity profiles describe what the platform provides. The mapping is not one-to-one.



Chips show the typical connectivity profiles, most common first; actual dependencies and requirements decide.

Regional flexibility (indicative)

Region-agnostic does not mean **region-random**.

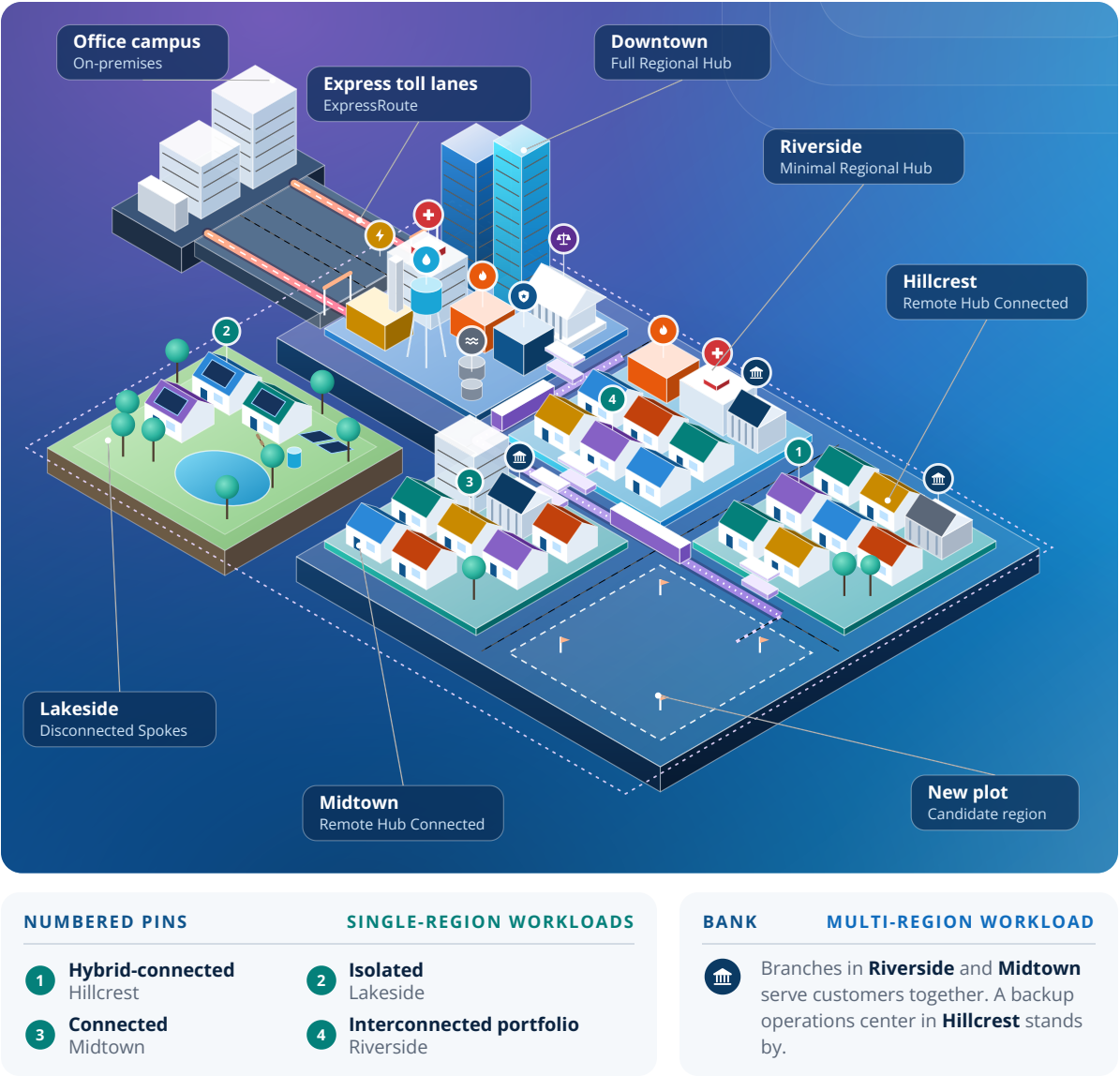
GOVERNANCE MOVES UP A LEVEL

Governance is not removed; it moves into policies that do not depend on particular regions. Instead of prescribing one regional architecture for every application, define a governed operating model and a small number of approved archetypes. Each application selects the **most efficient pattern** that satisfies its connectivity, resiliency, data, security, and service requirements.

THE MODEL AS A CITY

The platform is the city. Workloads are the people and businesses that live in it.

The city builds neighborhoods, services, roads, transit, and laws once. A family settles in one neighborhood; a bank spreads across several. Neither builds a new city.



PLACES

Neighborhood
An **Azure region**. The label names its connectivity profile.

Office campus
On-premises datacenters and users, outside the city.

Staked plot
A **candidate region**: zoned, but not built yet.

REGIONS AND ON-PREMISES

ROUTES

Express toll lanes
ExpressRoute. Two interchanges give the campus two separate paths into the city.

Transit line and stations
Private connectivity between regions. Hillcrest and Midtown ride it to Downtown.

CONNECTIVITY

LAWS

City hall
Laws, codes, and zoning: **governance and policy**, set once for the whole city.

City limits
Where the laws apply: every neighborhood, Lakeside included.

GOVERNANCE AND POLICY

CITY SERVICES

CITY SERVICES	SHARED SERVICES
Hospital or clinic	Fire station
Police station	Power plant
Water tower	Sewer plant

SHARED SERVICES

Together they are the platform's **shared services**. Downtown has every one. Riverside has a fire station and a clinic. Hillcrest and Midtown use Downtown's. Lakeside needs none.

FIGURE 7 The platform as a city. Each neighborhood is a region with its own connectivity profile. The legend explains each icon on the map.

06

GOVERNANCE MOVES UP A LEVEL · QUALIFY REGIONS

Regional qualification

Which Azure regions should be **prepared** to support workload requirements — not whether an individual workload should use multiple regions.

Regions enter the model through a repeatable assessment, not by assumption. Four checks decide which Azure regions should be prepared for which workload requirements, and each option is recorded with evidence, owners, and revalidation triggers. What the platform then provides in a qualified region — its connectivity profile, where each shared service is provided, and its hybrid connectivity — is selected next, and is not a qualification check. Whether an individual workload uses multiple regions remains a workload-specific design decision.

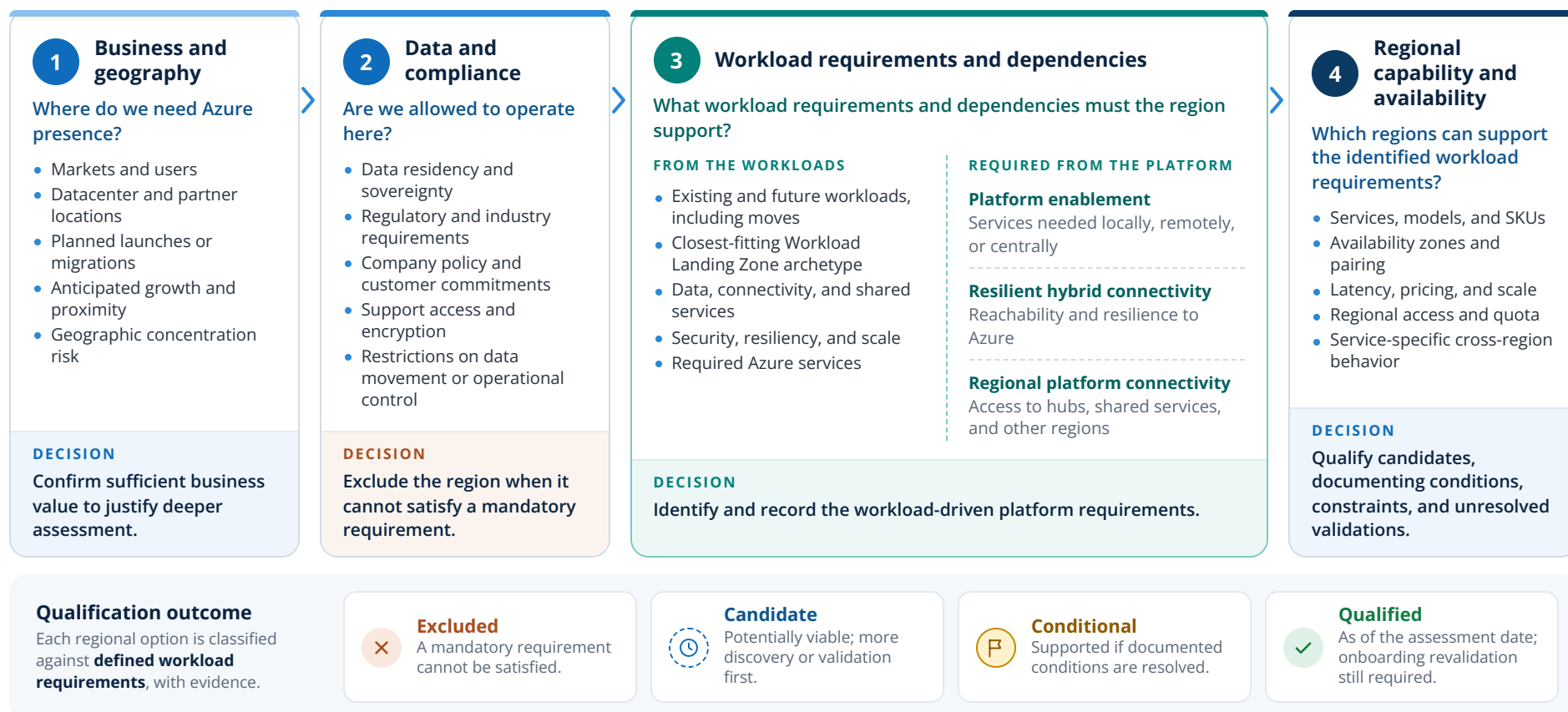


FIGURE 8 Four checks — the recommended sequence. Outcomes are specific to defined workload requirements, never a blanket certification; workload resiliency requirements are identified during workload design.

WHY THIS CREATES AGILITY

07 Region becomes a deployment parameter

Strategic hubs remain deliberate choices. Workloads stop inheriting those choices automatically.

TODAY

Region as a permanent architectural dependency

Every workload inherits the same regional pair and its full platform footprint, whether or not its requirements need it.

```
region = "primary" // fixed for every workload
```



WITH THIS MODEL

Region as a deployment parameter

Each workload selects from qualified regional options that fit its archetype and requirements, validated at decision time.

```
region = qualified_option(archetype, requirements)
```

WHEN CIRCUMSTANCES CHANGE



Capacity is available somewhere else

A qualified region with the required SKUs and quota can take the next deployment.



A required service is stronger in another region

AI models, services, or zone support can favor a different qualified option.



The regulatory or energy environment changes

Placement can adapt within the governed geography instead of waiting for a redesign.



More options for where the next workload can go

Existing workloads can stay put; new and movable ones use the best qualified option available.

Region A

Region B

Region C

Region E

+ qualified options

FIGURE 9 Regional optionality. Placement is revalidated at decision time for services, SKUs, AI models, regional access, and quota; capacity remains a deployment-time condition. Staying in the current region is always a valid outcome.



Deliberate where it matters: where the major strategic hubs live.



Adaptable where it helps: which regions individual applications consume.

08

AGILITY IN PRACTICE

Four scenarios at a glance

From a business or technical driver to a governed regional and workload decision. Illustrative, not customer-specific recommendations.

1 Regulated bank adopts a new in-country region

A new in-country region opens while regulation requires certain data or services to remain there.

QUALIFICATION Qualified for defined Hybrid-Connected requirements; other workloads remain candidates.

CONNECTIVITY Minimal Regional Hub; existing hybrid connectivity reused where it permits.

PLACEMENT In-scope workloads relocate selectively; others remain in place.

☞ Qualify for defined requirements without forcing portfolio-wide relocation; if the new region is nonpaired, design the recovery path explicitly.

2 AI team requires a specific model or GPU SKU

The model, SKU, or service is unavailable — or quota-limited — in existing regions.

QUALIFICATION Qualified for the defined AI requirements; conditional where private connectivity is not yet designed.

CONNECTIVITY Disconnected Spokes where self-contained; connected only where dependencies justify it.

PLACEMENT Net-new deployment, revalidated at decision time.

☞ Do not introduce enterprise connectivity or regional platform capabilities before a workload requirement justifies them.

3 Acquisition brings an estate in another geography

Interdependent applications, shared data, and local users where there is little Azure presence.

QUALIFICATION Candidate until dependencies are understood; then qualified for defined portfolio requirements.

CONNECTIVITY Remote Hub Connected, or Minimal Regional Hub where east-west traffic or independence justify it.

PLACEMENT Remain in place initially; relocate selectively as integration proceeds.

☞ Aggregate portfolio dependencies determine the profile; application count or acquisition status does not.

4 Datacenter exit accelerates migration

Hybrid-connected applications must leave a colocation facility; is another region now required?

QUALIFICATION Existing target region remains qualified; an additional region stays a candidate.

CONNECTIVITY No new profile: the existing strategic hub is reused; no new hub solely because another region is available.

PLACEMENT Most workloads land in the existing qualified region.

☞ Remaining in an existing qualified region is a valid outcome; a new region can stay an option, not a project.

09

THE STRATEGIC BENEFIT

Architect so that perfect prediction is unnecessary

No one can predict which European regions will offer the best combination of capacity, service availability, energy supply, and regulatory environment three to five years from now. **Architect so that it does not need to be predicted perfectly.**

That means separating what must stay consistent — governance, security, compliance, and the operating model — from what should stay adaptable: which regions individual applications consume.

For regulated organizations, application-level, region-agnostic placement is a way to de-risk cloud growth if future power restrictions affect the industry's ability to keep expanding hyperscale capacity in a particular country.



Lightweight Workload Landing Zone archetypes

Applications select the most efficient pattern that meets their requirements.



Geographically resilient connectivity

Diverse peering locations into the Azure backbone, planned for the geography.



A small number of strategic geographic hubs

The full footprint lives where it is justified, not in every region.

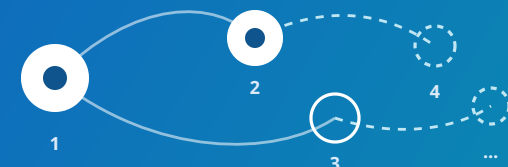


One governed operating model

Landing-zone guardrails and operating standards stay stable across the estate.

Together: an architecture that can evolve as Europe evolves.

The goal is not to select the perfect two regions. **It is to make additional regions significantly easier to adopt when business and workload requirements justify them.**



The adaptable **multi-region** Azure platform

An executive summary of the Multi-Region Platform Whitepaper. The full paper contains the detailed qualification checks, profile criteria, what to deploy for each profile, placement steps, and service-level resiliency guidance.

Multi-Region Platform · Executive Brief · October 2026

